

MODEL ANSWERS & MARKING SCHEME

Subject: Cyber Law and Ethics (SET-1)

Total Marks: 10 Marks

Question 1 (5 Marks)

A financial company notices that before a ransomware attack, several employees received suspicious emails and unknown login attempts were detected from foreign IP addresses. Analyze the possible pre-attack plans used by criminals. Identify and explain at least four stages involved in this attack preparation.

Four Stages of Pre-Attack Preparation:

- **1. Passive Reconnaissance:** Gathering publicly available information about the financial firm's domain, employee structure, and tech stack from social networks and public registers without tracing footprints.
- **2. Active Reconnaissance (Scanning):** Conducting unauthorized login attempts from foreign IPs to probe active firewalls, perimeter interfaces, and discover authentication gaps.
- **3. Weaponization & Phishing Setup:** Designing tailor-made malicious attachments or lookalike subdomains aimed precisely at internal staff profiles discovered during structural research.
- **4. Initial Access/Infiltration:** Dispatching unverified tracking or exploit delivery emails to secure one weak vector endpoint inside the domain node for ransomware spreading.

Question 2 (5 Marks)

A hacker collected employee details from LinkedIn and the company website before launching a phishing campaign. Analyze this scenario from the perspective of Cyber Law and Ethics. Identify the legal boundaries of Open Source Intelligence (OSINT) and the ethical implications of targeting individual employees.

Legal and Ethical Analysis:

- **Legal Status of OSINT:** Compiling open directories like LinkedIn remains lawful by configuration since users upload data willfully. However, scraping at scale violates user service conditions and local data protection regulations (e.g. processing identifier matrices without operational consent). Using that harvested database to dispatch malicious exploit campaigns directly establishes criminal intent, violating IT Acts.

- **Ethical Implications:** Weaponizing public registries violates individual trust parameters. Threat actors use deceptive framing parameters to trigger psychological compliance, offloading extreme network exploitation penalties onto regular employees.